

What are Constraints in MySQL?

Constraints are rules applied to table columns to ensure the **accuracy, reliability, and integrity** of the data stored in a database.

Types of Constraints in MySQL

Constraint	Description
NOT NULL	Ensures that a column cannot have a NULL value.
UNIQUE	Ensures all values in a column are unique.
PRIMARY KEY	Uniquely identifies each record in a table (combines NOT NULL + UNIQUE).
FOREIGN KEY	Maintains referential integrity between two tables.
CHECK	Ensures that values in a column meet a specific condition.
DEFAULT	Provides a default value for a column when none is specified.

Example Table Creation

Let's create two tables:

- students
- courses

1. Create courses Table

```
CREATE TABLE courses (  
  course_id INT PRIMARY KEY,  
  course_name VARCHAR(50) NOT NULL,  
  duration INT CHECK (duration > 0)  
);
```

Explanation:

- course_id → PRIMARY KEY (unique & not null)
- course_name → cannot be NULL
- duration → must be greater than 0 (CHECK constraint)

Insert Data:

```
INSERT INTO courses VALUES  
(101, 'MySQL Basics', 30),  
(102, 'Python Programming', 40);
```

2. Create students Table

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    age INT CHECK (age >= 18),  
    course_id INT,  
    admission_date DATE DEFAULT (CURRENT_DATE),  
    FOREIGN KEY (course_id) REFERENCES courses(course_id)  
);
```

Explanation:

- student_id → uniquely identifies student (PRIMARY KEY)
 - name → cannot be NULL (NOT NULL)
 - email → no duplicates allowed (UNIQUE)
 - age → must be 18 or above (CHECK)
 - course_id → linked to courses table (FOREIGN KEY)
 - admission_date → automatically gets current date if not provided (DEFAULT)
-

Insert Data:

```
INSERT INTO students (student_id, name, email, age, course_id)  
VALUES  
(1, 'John Doe', 'john@example.com', 22, 101),  
(2, 'Sara Khan', 'sara@example.com', 24, 102);
```

Example of Violating a Constraint:

-- This will fail because age < 18 (CHECK constraint violated)

INSERT INTO students VALUES (3, 'Mike', 'mike@example.com', 17, 101, '2025-10-06');

View Data

SELECT * FROM courses;

SELECT * FROM students;

Summary Table

Constraint	Example	Description
NOT NULL	name VARCHAR(50) NOT NULL	Prevents null values
UNIQUE	email VARCHAR(100) UNIQUE	Ensures no duplicate values
PRIMARY KEY	student_id INT PRIMARY KEY	Unique identifier for records
FOREIGN KEY	FOREIGN KEY (course_id) REFERENCES courses(course_id)	Ensures valid link to another table
CHECK	CHECK (age >= 18)	Ensures a condition is true
DEFAULT	admission_date DATE DEFAULT (CURRENT_DATE)	Sets default value automatically